# Curriculum Vitae

## Name and Contact Details

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| Title | Prof. |
| Name and Surname | Emile Rugamika CHIMUSA |
| Department | Applied Sciences |
| Institution | Northumbria University |
| Current position | Professor of Bioinformatics and Computational Biology |
| Home Address | 18 Beatrice Road, Heaton, Newcastle Upon Tyne, NE6 5RJ, UK |
| Phone | +44 7466147499 |
| Emails | echimusa@gmail.com |
| Github | https://github.com/echimusa |
| Website from the current position | https://www.northumbria.ac.uk/about-us/our-staff/c/emile-rugamika-chimusa/ |

## Career Profile (Education and Employment)

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| 2022 - present | Professor of Bioinformatics and Computational Biology, Northumbria University Newcastle, UK. |
| 2024 | Senior Fellowship, UK Professional Standards Framework for teaching and learning support in higher education. |
| 2020 – 2022 | Programme Director of Computational Health Informatics (CHIP), Faculty of Health Sciences, University of Cape Town, South Africa. (Associate Professor) |
| 2020 – Present | Founder of Omics Data Solutions Limited, Ireland. |
| 2019 – 2022 | Associate Professor of Computational Biology, Division of Human Genetics, Department of Pathology, University of Cape Town, Cape Town South Africa. |
| 2016 - 2022 | Programme convener of Medical Sciences in Honours in Human Genetics & Forensic Genetics, Division of Human Genetics, Department of Pathology, University of Cape Town, Cape Town South Africa. |
| 2016 - Present | PI and Bioinformatics ecosystem, High-Performance Computing research platform, Centre for High-Performance Computing (CHPC), Cape Town, South Africa. |
| 2016 – 2019 | Senior Lecturer, Division of Human Genetics, Department of Pathology, University of Cape Town, Cape Town South Africa. |
| 2014 – 2016 | Lecturer, Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, Cape Town, South Africa. |
| 2013 – 2014 | Postdoctoral researcher fellow, Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, Cape Town, South Africa. |
| 2013 – 2014 | Computational Biostatistic Scientific Researcher; Centre for Proteomic and Genomic research (CPGR), Cape Town, South Africa. |
| 2013 - | Invited lecturer of Statistical Genomics with R/Python programming Languages, African Institute for Mathematical Sciences, South Africa, Kigali, and in Ghana. |
| 2009 – 2013 | Ph.D. (Awarded in 2013) in Computational Biology<br>**Title**: *Mapping genes underlying ethnic differences in tuberculosis risk by linkage disequilibrium in the South African Coloured population of the Western Cape (https://open.uct.ac.za/handle/11427/10691 ).*<br>Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, South Africa and Visiting PhD student at Harvard T.H. Chan School of Public Health, USA.<br>**Supervisors:** Profs. Nicola Mulder, Alkes Price and Van Helden, Eileen Hoal |

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| 2011 - 2012 | Part-time lecturer, African Institute for Mathematical Sciences (AIMS), Cape Town, South Africa |
| 2008 – 2009 | **Masters** (Upgraded to PhD in 2009) in Computational Biology<br>Division of Computational Biology, Department of Integrative Biomedical Sciences, University of Cape Town, South Africa |
| 2007 – 2008 | Postgraduate diploma. (Awarded in 2008) Mathematical Sciences<br>University of Western Cape, African Institute for Mathematical Sciences (AIMS), University of Western Cape, Cape Town South Africa |
| 2005 – 2007 | PostgreSQL database administrator; Department of Information Technology/United Nations Mission DR Congo (MONUC) Independent Electoral Commission (CEI), Kinshasa |
| 1999 – 2005 | Bachelor Honours (Awarded in 2005, First Class) Applied Mathematics and Informatics. University of Kinshasa, Democratic Republic of Congo. |

## History of Mentoring and Supervision

| Name | Position | Year of graduation / completion | Current role & location |
|---|---|---|---|
| Francis Agamah | PhD Student | 2024 | Postdoctoral Research fellow, University of Cape Town, Computational Biology |
| Jacquiline Wangui Mugo | PhD Student | 2023 | Postdoctoral Research fellow at Allergology and Clinical Immunology, Department of Medicine, University of Cape Town Lung Institute |
| Abdoulaye Yalcouye | PhD Student | 2022 | Postdoctoral Research Fellow at the Johns Hopkins University School of Medicine |
| Edmond Tingang Wonkam | PhD Student | 2021 | Postdoctoral Research Fellow at The National Institutes of Health (NIH)<br>Bethesda, Maryland, United States |
| Ephifania Geza | PhD Student | 2021 | Data Analysist at Computation Biology Unit, University of Cape Town, South Africa |
| Lerato Majara | PhD Student | 2021 | Postdoctoral Research fellow at Dept of Psychiatry and Mental Health at the University of Cape Town |
| Delesa Damena Mulisa | PhD Student | 2020 | Postdoctoral Research fellow at The National Institutes of Health (NIH), Bethesda, Maryland, United States |
| Denis Awany | PhD Student | 2020 | Postdoctoral research fellow at the South African Tuberculosis Vaccine Initiative (SATVI), University of Cape Town |
| Khuthala Mnika | PhD Student | 2020 | Senior Medical Scientist at National Health Laboratory Service<br>University of Cape Town |

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| | | | City of Cape Town, Western Cape, South Africa |
| Prisca Kerapetse Thami | PhD student | 2020 | Postdoctoral Research Fellow · Africa Health Research Institute (AHRI), University of KwaZulu-Natal, South Africa · |
| Samar Elsheikh | PhD Student | 2019 | Postdoctoral Research Fellow at The Centre for Addiction and Mental Health, Toronto Canada |
| Nyarai Soko | PhD Student | 2017 | Postdoctoral Researcher - Ocular genomics<br>University of Cape Town<br>City of Cape Town, Western Cape, South Africa |
| Christian Domilongo Bope | Postdoctoral researcher | 2017 | Assoc Professor at Department of Mathematics & Computer Sciences, Faculty of Sciences at University of Kinshasa |
| Etienne Ntumba Kabongo | MSc Student | 2020 | PhD at UFR Médecine et Techniques Médicales, Université de Nantes |
| Francis Agamah | MSc Student | 2019 | PhD at the University of Cape Town, South Africa. |
| Peter Opiyo Kimathi | MSc Student | 2019 | Bioinformatics Analyst at the Centre of Proteomics and Genomics Research, South Africa |
| Shatha Alosaimi | MSc Student | 2019 | PhD at the King Saud Bin Abdulaziz University for Health Sciences, Saudi Arabia |
| Noluthando Rearabetswe Manyisa | MSc Student | 2018 | PhD at the University of Cape Town, Western Cape, South Africa |
| Amy Geard | MSc Student | 2017 | Postdoctoral Research Associate at London's global university |
| Joel Defo | MSs Student | 2016 | PhD at the University of Cape Town, South Africa |
| Milaine Tchamga | MSc Student | 2015 | Postdoctoral researcher at African Institute for Mathematical Sciences, Stellenbosch University. |
| Samson Wangila Wanyonyi | MSc Student | 2015 | PhD at the University of Eldoret, Kenya |
| Chacha Marwa Issarow | MSc student | 2015 | Research officer in Bioinformatics, University of Cape Town, South Africa |
| Nadia Solime Taou | MSc Student | 2014 | Research Associate, School of Social Sciences, Edinburgh Business School, UK |
| Abdoelnaser Mahmoud Degoot | MSc Student | 2014 | PhD and Lecturer at the University of Khartoum |

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| Rosemary Akinyi Aogo | MSc Student | 2014 | Data Scientist and Research at The National Institutes of Health, Bethesda, Maryland, USA |

## Details of Relevant Research Funding as Applicant Lead/Co

| **Grant Title** | **Agency** | **Amount / Period** | **Role / Funding Number** |
|---|---|---|---|
| Academy of Medical Sciences, UK | Academy of Medical Sciences Professorship Scheme, UK | £1,500,000 **2023 - 2026** | Academy of Medical Sciences, UK |
| Computational Omics and Biomedical Informatics Program (COBIP) | The National Institutes of Health (NIH), US | $ 1,730,785 **2021 - 2026** | PI 1U2RTW012131-01 |
| Developing a Sickle Africa Data Coordinating Center (SADaCC). | The National Institutes of Health (NIH)/NHGRI | $3,577,904 **2021-2025** | Expert/ Consultant U24HL135600-02 |
| Novel anticancer compounds targeting unusual nucleic acid structures in cancer therapy: Synthesis, biological evaluation and in silico approaches | Brazil, Russia, India, China, and South Africa (BRICS) multilateral projects - The South African Medical Research Council | ZAR 1,500,000 **2021 - 2023** | PI RG0034689 |
| H3ABioNet - Pan African Bioinformatics Network | The National Institutes of Health (NIH)/NHGRI | $2, 500,000 **2017 - 2023** | Co-PI U01HG009716 |
| Trusted World of Corona (TWOC) - PhD scholarship and joint supervision's contract between UCT and Radboud University, Nijmegen, Netherlands Recipient: Francis Agamah | Health~Holland | **€110,000 (approx.) 2021 - 2023** | PI PRJ001483 |
| Pharmacogenomics of tamoxifen in the treatment of breast cancer among South Africans | UCT ACC | ZAR 1,900,000. 2019 - 2022 | Co-PI UCT0017689 |
| Centre for High Performance Computing (CHPC) National Flagship Project - Computing Ecosystems C6100 equipment award | The Council for Scientific and Industrial Research (CSIR) South Africa | ZAR 2,850,000 2018 | PI SRC0027681 |
| Hearing Impairment Genetics Studies in Africa (HI-GENES Africa) | Wellcome Trust/AESA | $2,067,742 2017 - 2022 | Co-investigator H3A/18/001 |
| Hearing Impairment Genetics Studies in Africa (HI-GENES Africa) | The National Institutes of Health (NIH)/NHGRI | $1,249,097 2017 - 2022 | Co-PI U01HG009716 |

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| Individual Findings in Genetic Research in Africa (IFGeneRA) | The National Institutes of Health (NIH)/NHGRI | $2,499,081<br>2017-2021 | Co-investigator<br>1U54HG009790-01 |
| Computational method to understand population admixtures. | The International Development Research Centre (IDRC) - Canada | $120,000<br>2014-2018 | PI<br>AASRG/<br>SAMCD2008016S |

## Innovation/ Commercialization activity

2020 - Promoter and founder of Omics Data Solutions Limited, Dublin, Ireland.
2017-2018 Bioinformatics Expert for H3Africa consortium in designing first African custom genotyping/sequencing array Chip, H3Africa chip.
2014 – 2018 consulted for The Centre for Proteomics and Genomics Research (CPGR on the Knowledge Transfer Programme (KTP) system, to efficiently train and transfer "omics" knowledge among researchers and faculty on the African continent (Chimusa et al. 2015 PLoS Comput Biol; PMID: 26583922).

## Other Information as Appropriate

I have published up to 90 peer-reviewed papers of which 98 % are published in international top 10% high impact factor journals, 3 chapters in Books, and 1 published book in the field of Bioinformatics, genetics epidemiology, and medical population genetics.
> 3,606 citations; h-index: 32 (Google Scholar, Dec 2021).
Research primarily in bioinformatics method tools pertinent to genetic and environmental determinants of susceptibility to complex diseases, genome variation, microbial genomics, and infectious diseases from both the host and pathogen perspectives to facilitate the transformation of genomics-driven clinical practice.

I developed various genomics tools for post and functional genome-wide association (such as ancGWAS, PROXYANC, ancMETA), risk prediction, ancestry inference and genomics simulations (FractalSIM and FRANC), Gene Ontology semantic similarity-based functional analysis tool (A-DaGO-Fun) that are widely in the genomics and Bioinformatic research communities (https://github.com/echimusa).

My research group has a strong track record of combining original bioinformatics research with collaborative analysis of genomics datasets, facilitating access by collaborating experimental groups to new genomics techniques and data sets.

## Other Experience and Key Practical Skills

- Advanced Artificial Intelligence (AI), Machine Learning (ML) and Deep Learning
- Advanced software tool development with Python and R programming languages
- Computational Research & Development
- Classification Algorithms
- Relational database with SQL and PostgreSQL
- Parallel and edge Computing and shell scripting (Linux systems)
- Possessed technical skills on HTML, CSS, JQuery, JSON, XML.

## Professional Memberships

2010 - Member, International Society for Computational Biology (ISCB)
2011 - Member, Southern African Society for Human Genetics (SASHG)
2013 - Member/Co-I, Human Heredity and Health in Africa (H3Africa)

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| 2013 - | Member/Co-PI, Human Heredity and Health in Africa (H3ABionet) |
| 2013 - | Member, International Society for Computational Biology (ISCB) |
| 2013 - | Member, European Conference on Computational Biology (ECCB) |
| 2014 - | Member and Public relation, South African Society for Bioinformatics (SASBi) |
| 2017 - | Associate Editorial Board Member, Bentham Science |
| 2018 - | Associate Member, Institute for Infectious Disease and Molecular Medicine (IDM) |
| 2019 – 2021 | Guest editorial board of Bioinformatics and Briefings in Bioinformatics |
| 2019 - 2021 | Member of Editorial Board, Frontiers in Oncology |
| 2021 - | Co-chair for State of Data Science Collection in the Nature portfolio of journals |
| 2022- | Scientific editorial board of PLoS One |
| 2022 - | North of England Genetic Epidemiology Group (NEGEG), UK |
| 2022 - | Data Science for Health Discovery and Innovation in Africa (DS-I Africa) Initiative |
| 2023 - | Northumbria Turing Universities Network, UK |

## Honors

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| 2010 - 2010 | Prize, Intel Software Multi-core and Parallel programming, Stellenbosch University |
| 2011 - 2011 | Young Investigator Awards, 12th International Congress of Human Genetics and The American Society of Human Genetics, Young Investigator Awards |
| 2012 - 2014 | Carnegie Corporation of New York, PhD Scholarship and Postdoctoral award, University of Cape Town |
| 2014 - 2014 | Summer school in Advanced Monte Carlo Methods for Complex Inference Problems., Isaac Newton Institute for Mathematical Sciences, Cambridge, UK |
| 2017 | Awarded C2 Rating, National Research Foundation (NRF), South Africa |
| 2017 | Elected as an Affiliate of The World Academy of Sciences (TWAS), Japan |
| 2018 | Elected as Future Leader, Science and Technology in Society (STS) forum |

## Additional relevant activities

- 2021 - Grant reviewer for Wellcome Trust,
- 2020 - Grant reviewer for Genome Canada,
- 2020 – Expert Grant reviewer EU HORIZON
- 2022 - 2022 Grant reviewer National Institute for Health and Care Research
- 2016 - 2021 Grant reviewer for the South African National Research Foundation
- 2020 – 2021 Grant reviewer for The African Academy of Sciences.
- 2021 - Recently appointed by Fogarty International Center of the National Institutes of Health (NIH) as scientific co-chair for a collection of scientific papers to be published in the Nature portfolio of journals on the state of the field of data science for health in Africa.
- Frequently Invited reviewer of various international journals including The American Journal of Human Genetics, Briefings in Bioinformatics, Bioinformatics, Scientific Reports, PLoS Computational Biology

## Selected International Workshop/Conference Organised

| Name | Place | Size | Type | Date | role |
|---|---|---|---|---|---|
| African symposium and 16th tyan international thematic virtual workshop on big data, analytics and machine intelligence | akure, Nigeria, African symposium and 16th tyan international thematic virtual workshop on big data, analytics and machine intelligence | 46 | Symposium and Workshop | 24/06/2024 | Speaker & Organizer |

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| AI and Data Science in Multi-Omics for Human Health Workshop | Newcastle, UK https://www.northumbria.ac.uk/about-us/news-events/events/2023/07/ai-and-data-science-in-multi-omics-for-human-health-workshop/ | 82 | Symposium and Workshop | 27-28 July 2023 | Organiser and speaker |
| 2nd African Symposium and 6th TYAN International Thematic Virtual Workshop on Big Data, Analytics and Machine Intelligence | Nigeria https://tyan.twas.org/events/2nd-african-symposium-and-3rd-tyan-international-thematic-virtual-workshop-on-big-data-analytics-and-machine-intelligence/ | 88 | Symposium and Workshop | 3-4 Dec 2020 | Co/organizer, Keynote Speaker, and Trainer |
| 4th Tyan International Thematic Workshop and 1st African Symposium on "Big Data, Analytics and Machine Intelligence for Financial, Health and Environmental Inclusion in Developing Countries" | Akure, Nigeria http://tyan.twas.org/event/currentevent | 67 | Symposium and Workshop | 10-14 June 2019 | Co/organizer, Keynote Speaker, and Trainer |
| Leveraging the Power of High-Performance Computing for Genotype Calling and Next Generation Sequencing Data Analysis: Prioritizing Mutations and Enrichment Analysis | University of Cape Town, Cape Town South Africa https://www.sickleinafrica.org/node/314 | 47 | Workshop | 17-21 June 2019 | Organizer and main Trainer |
| Big Genomics Data Analysis | H3Africa Consortium Kigali, Rwanda. | 19 | Workshop | 16-17 Sept 2018 | Invited trainer |
| Introduction to Genome-Wide Association Studies (GWAS) in Complex Disease Workshop | AIMS and University of Ghana http://aimsgh-bioinformatics.weebly.com/ | 56 | Workshop | 8-20, June 2017 | Co-organizer and Main trainer |
| Introduction to Genomics Big Data Sciences Workshop | AIMS Ghana http://aimsgh-bioinformatics2.weebly.com/ | 37 | Workshop | 8-16 July, 2017 | Co-organizer and Main trainer |
| African Symposium on Genome Wide Association Studies (ASGWAS) for complex disease Conference | False Bay College, Muizenberg https://www.h3africa.org/component/content/article/9-news/212-1st-annual-african-symposium-on-genome-wide-association-studies-for-complex-disease-23rd-24th-april-2015 | 138 | Conference | 23-24 April 2015 | Co-founder, Organizer, share session, chair panel discuss |
| RNA-Seq data analysis with Chipster. Centre for high Performance Computing, Cape Town, South Africa. | www.chpc.ac.za | 26 | Workshop | 2015 | Invited trainer |

## Selected international conferences

| Reference | Location | Invitation Oral/poster |
|---|---|---|
| **1. Emile R. Chimusa (2024),** Affordable AI-driven Wearable Biosensing network for early detection and self-monitoring. *International Healthcare Facilities Management Symposium, Türkiye 2024.* | Virtual | Keynote |
| **2. Emile R. Chimusa** (2021) Next-generation of disease risk prediction and risk stratification. *The 6th TACC annual meeting, 2023.* | Virtual | Keynote Speaker |
| **3. Emile R. Chimusa** (2021) Tantalizing Dilemma in Genetics Risk Prediction: Case of African Genomics. *The Curating the Clinical Genome conference, Wellcome Genome Campus Advanced Courses, and Scientific Conferences programme.* | UK, London 2021 (Virtual) | Keynote Speaker |
| *4.* **Emile R. Chimusa** (2021) Tantalizing Dilemma in Diseases scoring statistics. *Data Science for Solution-Driven and Sustainable Response to Current Challenges in the Developing World. 2nd African Symposium and 6th TYAN International Thematic Virtual Workshop on Big Data, Analytics, and Machine Intelligence by TWAS.* | Trieste, Italy, 2020 (Virtual) | Keynote Speaker |
| **5. Emile R. Chimusa** (2018) Bioinformatics in the Era of Post Genomics and Big Data. *TWAS 14th General Conference & 28th General Meeting, Trieste, Italy 27-29 November 2018.* | Trieste, Italy, 2018 | Invited Speaker |
| **6. Emile R. Chimusa** (2018) Post Genome-wide Association Analysis: Genes and Sub-Networks Underlying Complex Disease Risk. *The Future Readers Program, the 15th Annual Meeting of Science and Technology in Society forum (STS forum) October 2018 Kyoto, Japan.* | Kyoto, Japan. 2018 | Oral |
| **7. Chimusa, E.R***., Mbiyavanga, M., Mazandu, G.K. and Mulder, N.J. ancGWAS: a post genome-wide association study method for interaction, pathway and ancestry analysis in homogeneous and admixed populations. *10th Annual World Cancer Congress (WCC-2017), Barcelona, Spain.* | Barcelona, Spain, 2017 | Invited speaker |
| 8. **Emile R. Chimusa** (2017) TagMix: An integrated Genome-Wide Tag SNPs Selection models for Custom Chip design in multiple populations of high genetic variation and low LD. *Washington DC, USA, October 11-14, Seventh H3Africa Consortium Meeting.* | Washington DC, USA | poster |
| *9.* **Emile R. Chimusa** (2015) TagMix: An integrated Genome-Wide Tag SNPs Selection models for Custom Chip design in multiple populations of high genetic variation and low linkage disequilibrium. *Intelligent Systems for Molecular Biology 2017 and 14th European Conference on Computational Biology. Dublin, Ireland.* | Dublin, Ireland, 2016 | Invited speaker |
| 10. **Emile R. Chimusa** (2014) Leveraging cross-population Gene/Sub-network Meta-Analysis to recover signal underlying ethnic difference in disease. *22st Annual International Conference on Intelligent Systems for Molecular Biology (ISMB), Boston, USA.* | Boston, USA, 2014 | Oral |

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| 11. **Emile R. Chimusa** (2014) A probabilistic approach for modelling natural selection in pinpointing ancestry along the genome of an individual of mixed ancestry, Monte Carlo Inference for High-Dimensional Statistical Models. *Monte Carlo Inference for High-Dimensional Statistical Models, Cambridge UK.* | Cambridge UK | speaker |
| 12. **Emile R. Chimusa** (2014) Leveraging ancestry along the genome of admixed individuals to resolve missing heritability in disease scoring statistics. *13th European Conference on Computational Biology, France*. | France, 2014 | Invited speaker |
| 13. **Emile R. Chimusa** (2014) Graph-based Approach for Analysing Sub-Networks Underlying Ethnic Differences in Complex Disease Risk in Admixed Populations. *21st Annual International Conference on Intelligent Systems for Molecular Biology (ISCB) and 12th Annual European Conference on Computational Biology. Berlin, Germany.* | Berlin, Germany, 2014 | speaker |
| **14. Emile R. Chimusa** (2013) efficient algorithms for selecting the best proxy ancestry in recently admixed populations: Assessment of population bottleneck and founder effect in the Coloured population. *20st European Conference on Computational Biology (ECCB) Basel, Switzerland.* | Basel, Switzerland, 2013 | Speaker |
| 15. **Emile R. Chimusa** (2013) Genome-Wide Association Study of TB in the South African Coloured Population: Comprehensive Gene and Pathway-Based Association Study. *12th International Congress of Human Genetics and the American Society of Human Genetics. Montreal, Canada.* | Montreal, Canada, 2013 | Speaker |

## Selected Publications

See full list include 2024 at **https://orcid.org/0000-0001-8846-2047**

| Article | IF |
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### 2024

1. Ginete, C., Delgadinho, M., Santos, B., Miranda, A., Silva, C., Guerreiro, P., Chimusa, E.R. and Brito, M., 2024. Genetic Modifiers of Sickle Cell Anemia Phenotype in a Cohort of Angolan Children. Genes, 15(4), p.469.

### 2023

2. Katsukunya, J.N., Soko, N.D., Naidoo, J., Rayner, B., Blom, D., Sinxadi, P., **Chimusa, E.R.,** Dandara, M., Dzobo, K., Jones, E. and Dandara, C., 2023. Pharmacogenomics of Hypertension in Africa: Paving the Way for a Pharmacogenetic-Based Approach for the Treatment of Hypertension in Africans. International Journal of Hypertension, 2023. **2.3**
3. Esoh, K.K., Apinjoh, T.O., Amambua-Ngwa, A., Nyanjom, S.G., **Chimusa, E.R**., Amenga-Etego, L., Wonkam, A. and Achidi, E.A., 2023. Genome-wide association study identifies novel candidate malaria resistance genes in Cameroon. *Human Molecular Genetics*, *32*(12), pp.1946-1958. **5.1**
4. Kumuthini J, Zass L., Chaouch M., Gill Z., Ras V., Mungloo-Dilmohamud Z., Dassen Sathan, Anisah Ghoorah, Faisal Fadlelmola, Christopher Fields, John Van Horn, Fouzia Radouani, Melissa Konopko, **Emile R. Chimusa**, Shakuntala Baichoo, 2023. Data standardization in the omics field. In *Genomic Data Sharing* (pp. 137-155). Academic Press.

5. Nembaware, V., Bennett, D., **Chimusa, E.R**., Chikowore, T., Daodu, R., Bitoungui, V.N., Williams, S.M., Fatumo, S., Healy, S., Seoighe, C. and Wonkam, A., 2023. The African Society of Human Genetics successfully launches global data science workshops. *Trends in Genetics*. **11.4**
6. Thami, P.K., Choga, W.T., Dandara, C., O'Brien, S.J., Essex, M., Gaseitsiwe, S. and **Chimusa, E.R**., 2023. Whole genome sequencing reveals population diversity and variation in HIV-1 specific host genes. *Frontiers in Genetics*, *14*, p.1290624.

## 2022

7. **Chimusa, E.R**., Alosaimi, S. and Bope, C.D., 2022. Dissecting generalizability and actionability of disease-associated genes from 20 worldwide ethnolinguistic cultural groups. Frontiers in Genetics, 13, p.835713.
8. Feldmann, D., Bope, C.D., Patricios, J., **Chimusa, E.R***., Collins, M. and September, A.V., 2022. A whole genome sequencing approach to anterior cruciate ligament rupture–a twin study in two unrelated families. PloS one, 17(10), p.e0274354.
9. **Chimusa ER***, Defo J. Dissecting Meta-Analysis in GWAS Era: Bayesian Framework for Gene/Subnetwork-Specific Meta-Analysis. Front Genet. 2022 May 18;13:838518. doi: 10.3389/fgene.2022.838518. PMID: 35664319; PMCID: PMC9159898.
10. Agamah, F.E., Bayjanov, J.R., Niehues, A., Njoku, K.F., Skelton, M., Mazandu, G.K., Ederveen, T.H., Mulder, N., **Chimusa, E.R**. and t Hoen, P.A., 2022. Computational approaches for network-based integrative multi-omics analysis. Frontiers in Molecular Biosciences, 9, p.1214.

## 2021

1. Tolba, M.F., Santa-Maria, C.A., Albini, A., Chimusa, E.R., Al-Ramadi, B.K. and Tolaney, S.M., 2021. Immunotherapy as an Evolving Approach for the Treatment of Breast Cancer. Frontiers in oncology, 11, p.752689. IF **6.244**
2. Elsheikh, S.S*., Chimusa, E.R*, Mulder, N. and Crimi, A., 2021. Relating Global and Local Connectome Changes to Dementia and Targeted Gene Expression in Alzheimer's Disease. Frontiers in Human Neuroscience,15 p.771, 1662-5161. **3.15**
3. Muyambo S, Ndadza A, Soko ND, Kruger B, Kadzirange G, Chimusa E.R, Masimirembwa CM, Ntsekhe M, Nhachi CFB, Dandara C. Warfarin Pharmacogenomics for Precision Medicine in Real-Life Clinical Practice in Southern Africa: Harnessing 73 Variants in 29 Pharmacogenes. OMICS. 2021 Dec 24. doi: 10.1089/omi.2021.0199. Epub ahead of print. PMID: 34958284. **2.089**
4. Damena, D., Agamah, F.E., Kimathi, P.O., Kabongo, N.E., Girma, H., Choga, W.T., Golassa, L. and **Chimusa, E.R***., 2021. Insilico Functional Analysis of Genome-Wide Dataset From 17,000 Individuals Identifies Candidate Malaria Resistance Genes Enriched in Malaria Pathogenic Pathways. *Frontiers in genetics*, *12*, p.676960. **3.789**
5. Choudhury A, Aron S, Botigué LR, Sengupta D, Botha G, Bensellak T, Wells G, Kumuthini J, Shriner D, Fakim YJ, Ghoorah AW, Dareng E, Odia T, Falola O, Adebiyi E, Hazelhurst S, Mazandu G, Nyangiri OA, Mbiyavanga M, Benkahla A, Kassim SK, Mulder N, Adebamowo SN, **Chimusa ER**, Muzny D, Metcalf G, Gibbs RA; TrypanoGEN Research Group, Rotimi C, Ramsay M; H3Africa Consortium, Adeyemo AA, Lombard Z, Hanchard NA. Author Correction: High-depth African genomes inform human migration and health. Nature. 2021 Apr;592(7856):E26. doi: 10.1038/s41586-021-03286-9. Erratum for: Nature. 2020 Oct;586(7831):741-748. PMID: 33846614; PMCID: PMC8081653. **49.96**

6. Agamah, F.E., Damena, D., Skelton, M., Ghansah, A., Mazandu, G.K. and **Chimusa, E.R***., 2021. Network-driven analysis of human–Plasmodium falciparum interactome: processes for malaria drug discovery and extracting in silico targets. *Malaria journal*, *20*(1), pp.1-20. **3.109**
7. Makinde, F.L., Tchamga, M.S., Jafali, J., Fatumo, S., Chimusa, E.R., Mulder, N. and Mazandu, G.K., 2021. Reviewing and assessing existing meta-analysis models and tools. *Briefings in Bioinformatics*, *22*(6), p.bbab324. **11.62**
8. Feldmann DC, Rahim M, Suijkerbuijk MAM, Laguette MN, Cieszczyk P, Ficek K, Huminska-Lisowska K, Häger CK, Stattin E, Nilsson KG, Alvarez-Rumero J, Eynon N, Feller J, Tirosh O, Posthumus M, **Chimusa ER**, Collins M, September AV. Investigation of multiple populations highlight VEGFA polymorphisms to modulate anterior cruciate ligament injury. J Orthop Res. 2021 Oct 18. doi: 10.1002/jor.25192. Epub ahead of print. PMID: 34664319. **3.14**
9. Mazandu, G.K., Opap, K., Makinde, F., Nembaware, V., Agamah, F., Bope, C., **Chimusa, E.R**., Wonkam, A. and Mulder, N.J., 2021. An Integrated Platform Supporting Semantic Similarity Score Calculation and Reproducibility. Research Square; 2021. DOI: 10.21203/rs.3.rs-806346/v1
10. Oluwole, O.G., Esoh, K.K., Wonkam-Tingang, E., Manyisa, N., Noubiap, J.J., **Chimusa, E.R***. and Wonkam, A., 2021. Whole exome sequencing identifies rare coding variants in novel human-mouse ortholog genes in African individuals diagnosed with non-syndromic hearing impairment. Experimental Biology and Medicine, 246(2), pp.197-206. **2.688**
11. Biobele J. Brown, Anazoeze Madu, Raphael Z. Sangeda, Siana Nkya, Emmanuel Peprah, Vivian Paintsil, Bruno P. Mmbando, Joyce Gyamfi, Chide E. Okocha, Samuel A. Asala, Victoria Nembaware, Mario Jonas, Andre P. Kengne, **Emile R. Chimusa**, Arthemon Nguweneza, Hezekiah A. Isa, Uche Nnebe-Agumadu, Adekunle D. Adekile, Alex Osei-Akoto, Kwaku Ohene-Frempong, Emmanuel Balandya, Obiageli E. Nnodu, Ambroise Wonkam, Julie Makani & The Sickleinafrica Consortium (2021) Utilization of Pneumococcal Vaccine and Penicillin Prophylaxis in Sickle Cell Disease in Three African Countries: Assessment among Healthcare Providers in SickleInAfrica, Hemoglobin, 45:3, 163-170, **0.848**
12. Hassan, R., Allali, I., Agamah, F.E., Elsheikh, S.S., Thomford, N.E., Dandara, C. and **Chimusa, E.R***., 2021. Drug response in association with pharmacogenomics and pharmacomicrobiomics: towards a better personalized medicine. Briefings in Bioinformatics, 22(4), p.bbaa292. **11.62**
13. Esoh, K.K., Apinjoh, T.O., Nyanjom, S.G., Wonkam, A., **Chimusa, E.R**., Amenga-Etego, L., Amambua-Ngwa, A. and Achidi, E.A., 2021. Fine scale human genetic structure in three regions of Cameroon reveals episodic diversifying selection. *Scientific reports*, *11*(1), pp.1-13. **4.379**
14. Ndadza, A., Muyambo, S., Mntla, P., Wonkam, A., **Chimusa, E**., Kengne, A.P., Ntsekhe, M. and Dandara, C., 2021. Profiling of warfarin pharmacokinetics-associated genetic variants: Black Africans portray unique genetic markers important for an African specific warfarin pharmacogenetics-dosing algorithm. *Journal of Thrombosis and Haemostasis*, *19*(12), pp.2957-2973. **4.662**
15. Alosaimi, S., van Biljon, N., Awany, D., Thami, P.K., Defo, J., Mugo, J.W., Bope, C.D., Mazandu, G.K., Mulder, N.J. and **Chimusa, E.R***., 2021. Simulation of African and non-African low and high coverage whole genome sequence data to assess variant calling approaches. *Briefings in Bioinformatics*, *22*(4), p.bbaa366. **11.62**
16. Mazandu, G.K., Hooper, C., Opap, K., Makinde, F., Nembaware, V., Thomford, N.E., **Chimusa, E.R.**, Wonkam, A. and Mulder, N.J., 2021. IHP-PING—generating integrated **11.62**

human protein–protein interaction networks on-the-fly. *Briefings in bioinformatics*, *22*(4), p.bbaa277.

## 2020

| | | |
|---|---|---|
| | | **6.15** |
| 17. | Wonkam, A*., **Chimusa, E.R***., Mnika, K., Pule, G.D., Ngo Bitoungui, V.J., Mulder, N., Shriner, D., Rotimi, C.N. and Adeyemo, A., 2020. Genetic modifiers of long-term survival in sickle cell anemia. *Clinical and Translational medicine*, *10*(4), p.e152. | **11.492** |
| 18. | Wonkam, A*., Manyisa, N., Bope, C.D., Dandara, C. and Chimusa, E.R*., 2020. Whole exome sequencing reveals pathogenic variants in MYO3A, MYO15A and COL9A3 and differential frequencies in ancestral alleles in hearing impairment genes among individuals from Cameroon. Human molecular genetics, 29(23), pp.3729-3743. | |
| 19. | Wonkam Tingang, E., Noubiap, J.J., F Fokouo, J.V., Oluwole, O.G., Nguefack, S., **Chimusa, E.R***. and Wonkam*, A., 2020. Hearing impairment overview in Africa: the case of Cameroon. *Genes*, *11*(2), p.233. | **3.688** |
| 20. | Awany, D., Allali, I. and **Chimusa, E.R***., 2020. Dissecting genome-wide studies for microbiome-related metabolic diseases. *Human Molecular Genetics*, *29*(R1), pp.R73-R80. | **6.15** |
| 21. | Awany, D. and **Chimusa, E.R***., 2021. Heritability jointly explained by host genotype and microbiome: will improve traits prediction? *Briefings in Bioinformatics*, *22*(3), p.bbaa175. | **11.62** |
| 22. | Damena, D. and **Chimusa, E.R*.,** 2020. Genome-wide heritability analysis of severe malaria resistance reveals evidence of polygenic inheritance. *Human molecular genetics*, *29*(1), pp.168-176. | |
| 23. | Nkya, S., Mwita, L., Mgaya, J., Kumburu, H., Van Zwetselaar, M., Menzel, S., Mazandu, G.K., Sangeda, R**., Chimusa, E**. and Makani, J., 2020. Identifying genetic variants and pathways associated with extreme levels of fetal hemoglobin in sickle cell disease in Tanzania. *BMC medical genetics*, *21*, pp.1-12. | **1.988** |
| 24. | Elsheikh, S.S., **Chimusa, E.R**., Mulder, N.J. and Crimi, A., 2020. Genome-wide association study of brain connectivity changes for alzheimer's disease. *Scientific reports*, *10*(1), pp.1-16. | **4.373** |
| 25. | | |
| 26. | Mazandu, G.K., Nembaware, V., Thomford, N.E., Bope, C., Ly, O., **Chimusa, E.R**. and Wonkam, A., 2020. A potential roadmap to overcome the current eastern DRC ebola virus disease outbreak: from a computational perspective. *Scientific African*, *7*, p.e00282. | **1.78** |

## 2019

| | | |
|---|---|---|
| 27. | Damena, D., Denis, A., Golassa, L. and **Chimusa, E.R***., 2019. Genome-wide association studies of severe P. falciparum malaria susceptibility: progress, pitfalls and prospects. *BMC medical genomics*, *12*(1), pp.1-14. | **2.93** |
| 28. | Thami, P.K. and **Chimusa, E.R***., 2019. Population structure and implications on the genetic architecture of HIV-1 phenotypes within Southern Africa. *Frontiers in genetics*, *10*, p.905. | **3.789** |
| 29. | Shatha Alosaimi, Armand Bandiang, Noelle van Biljon, Denis Awany, Prisca K Thami, Milaine S S Tchamga, Anmol Kiran, Olfa Messaoud, Radia Ismaeel Mohammed Hassan, Jacquiline Mugo, Azza Ahmed, Christian D Bope, Imane Allali, Gaston K Mazandu, Nicola J Mulder, **Emile R Chimusa***, 2020. A broad survey of DNA sequence data simulation tools. *Briefings in functional genomics*, *19*(1), pp.49-59. | **4.241** |
| 30. | Sickle Cell Disease Ontology Working Group **Chimusa, E**, 2019. The Sickle Cell Disease Ontology: enabling universal sickle cell-based knowledge representation. *Database*, *2019*. | **4.159** |

31. Wonkam, E.T., **Chimusa, E**., Noubiap, J.J., Adadey, S.M., Fokouo, J.V.F. and Wonkam, A., 2019. GJB2 and GJB6 mutations in hereditary recessive non-syndromic hearing impairment in Cameroon. *Genes*, *10*(11). **3.688**
32. Thomford, N.E., Bope, C.D., Agamah, F.E., Dzobo, K., Owusu Ateko, R., **Chimusa, E**., Mazandu, G.K., Ntumba, S.B., Dandara, C. and Wonkam, A., 2020. Implementing artificial intelligence and digital health in resource-limited settings? Top 10 lessons we learned in congenital heart defects and cardiology. *Omics: a journal of integrative biology*, *24*(5), pp.264-277. **3.24**
33. Geza, E., Mulder, N.J., **Chimusa, E.R***. and Mazandu, G.K*., 2020. FRANC: a unified framework for multi-way local ancestry deconvolution with high density SNP data. *Briefings in Bioinformatics*, *21*(5), pp.1837-1845. 13.99
34. Agamah, F.E., Mazandu, G.K., Hassan, R., Bope, C.D., Thomford, N.E., Ghansah, A. and **Chimusa, E.R***., 2020. Computational/in silico methods in drug target and lead prediction. *Briefings in bioinformatics*, *21*(5), pp.1663-1675. **13.99**
35. Mhandire, D., Duri, K., Kaba, M., Mhandire, K., Musarurwa, C., **Chimusa, E**., Munjoma, P., Mazengera, L., Stray-Pedersen, B. and Dandara, C., 2019. Seroprevalence of Cytomegalovirus infection among HIV-infected and HIV-uninfected pregnant women attending antenatal clinic in Harare, Zimbabwe. *Viral immunology*, *32*(7), pp.289-295. **7.561**
36. Damena, D., Tolosa, S., Hailemariam, M., Zewude, A., Worku, A., Mekonnen, B., Mohammed, T., Admasu, A., **Chimusa, E.R**., Mihret, A. and Abebe, T., 2019. Genetic diversity and drug susceptibility profiles of Mycobacterium tuberculosis obtained from Saint Peter's TB specialized Hospital, Ethiopia. *PloS one*, *14*(6), p.e0218545. **3.24**
37. Bope, C.D., **Chimusa, E.R**., Nembaware, V., Mazandu, G.K., de Vries, J. and Wonkam, A., 2019. Dissecting in silico mutation prediction of variants in African genomes: challenges and perspectives. *Frontiers in genetics*, *10*, p.601. **3.789**
38. Elsheikh, S.S.M., Bakas, S., Mulder, N.J., **Chimusa, E.R**., Davatzikos, C. and Crimi, A., Multi-Stage Association Analysis of Glioblastoma Gene Expressions with Texture and Spatial Patterns. Brainlesion: Glioma. *Multiple Sclerosis, Stroke and Traumatic Brain Injuries*, pp.239-250. **2.101**
39. **Chimusa, E.R***., Beighton, P., Kumuthini, J. and Ramesar, R.S., 2019. Detecting genetic modifiers of spondyloepimetaphyseal dysplasia with joint laxity in the Caucasian Afrikaner community. *Human molecular genetics*, *28*(7), pp.1053-1063.
40. Awany, D., Allali, I., Dalvie, S., Hemmings, S., Mwaikono, K.S., Thomford, N.E., Gomez, A., Mulder, N. and **Chimusa, E.R***., 2019. Host and microbiome genome-wide association studies: current state and challenges. *Frontiers in genetics*, *9*, p.637. **3.789**
41. Awany, D., Allali, I. and **Chimusa, E.R***., 2019. Tantalizing dilemma in risk prediction from disease scoring statistics. *Briefings in functional genomics*, *18*(4), pp.211-219. **4.241**
42. Davis, L., Beiter, E., Khramtsova, E., Van Der Merwe, C., **Chimusa, E**., Simonti, C., Stein, D., Capra, J., Knowles, J., Straub, P. and Hucks, D., 2019. Polygenic Adaptation Underlies Evolution of Brain Structures and Behavioral Traits. *European Neuropsychopharmacology*, *29*, pp.S755-S756. **4.6**
43. Mnika, K., Mazandu, G.K., Jonas, M., Pule, G.D., **Chimusa, E.R**., Hanchard, N.A. and Wonkam, A., 2019. Hydroxyurea-induced miRNA expression in sickle cell disease patients in Africa. *Frontiers in genetics*, *10*, p.509. **3.789**
44. Mufford, M., Cheung, J., Jahanshad, N., Van Der Merwe, C., Ding, L., Groenewold, N., Koen, N., **Chimusa, E.R**., Dalvie, S., Ramesar, R. and Knowles, J.A., 2019. Concordance of genetic variation that increases risk for tourette syndrome and that influences its underlying neurocircuitry. *Translational psychiatry*, *9*(1), pp.1-10. **6.22**

## 2018

45. **Chimusa, E.R*.**, Defo, J., Thami, P.K., Awany, D., Mulisa, D.D., Allali, I., Ghazal, H., Moussa, A. and Mazandu, G.K., 2020. Dating admixture events is unsolved problem in multi-way admixed populations. *Briefings in bioinformatics*, *21*(1), pp.144-155. **13.99**
46. **Chimusa, E.R***., Dalvie, S., Dandara, C., Wonkam, A. and Mazandu, G.K., 2019. Post genome-wide association analysis: dissecting computational pathway/network-based approaches. *Briefings in bioinformatics*, *20*(2), pp.690-700. **13.99**
47. Geza, E., Mugo, J., Mulder, N.J., Wonkam, A., **Chimusa, E.R**. and Mazandu, G.K., 2019. A comprehensive survey of models for dissecting local ancestry deconvolution in human genome. *Briefings in bioinformatics*, *20*(5), pp.1709-1724. **13.99**
48. Wonkam, A., Mnika, K., Ngo Bitoungui, V.J., Chetcha Chemegni, B., **Chimusa, E.R**., Dandara, C. and Kengne, A.P., 2018. Clinical and genetic factors are associated with pain and hospitalisation rates in sickle cell anaemia in Cameroon. *British journal of haematology*, *180*(1), pp.134-146. **5.67**
49. Thomford, N.E., Dzobo, K., Yao, N.A., **Chimusa, E**., Evans, J., Okai, E., Kruszka, P., Muenke, M., Awandare, G., Wonkam, A. and Dandara, C., 2018. Genomics and epigenomics of congenital heart defects: expert review and lessons learned in Africa. *Omics: a journal of integrative biology*, *22*(5), pp.301-321. **2.37**
50. Soko, N.D., **Chimusa, E**., Masimirembwa, C. and Dandara, C., 2019. An African-specific profile of pharmacogene variants for rosuvastatin plasma variability: limited role for SLCO1B1 c. 521T> C and ABCG2 c. 421A> C. *The pharmacogenomics journal*, *19*(3), pp.240-248. **3.55**
51. Elsheikh, S., **Chimusa, E.R**., Mulder, N. and Crimi, A., 2018, April. Relating connectivity changes in brain networks to genetic information in Alzheimer patients. In *2018 IEEE 15th International Symposium on Biomedical Imaging (ISBI 2018)* (pp. 1390-1393). IEEE.
52. Thomford, N.E., Dzobo, K., **Chimusa, E**., Andrae-Marobela, K., Chirikure, S., Wonkam, A. and Dandara, C., 2018. Personalized herbal medicine? A roadmap for convergence of herbal and precision medicine biomarker innovations. *Omics: a journal of integrative biology*, *22*(6), pp.375-391. **2.37**
53. Wanyonyi, S.W., Mbete, D.A. and **Chimusa, E.R***., 2018. Computational generalization of mixed models on large-scale data with applications to genetic studies. *Asian Journal of Probability and Statistics*, pp.1-31. **0.2**
54. Ndadza, A., Cindi, Z., Makambwa, E., **Chimusa, E**., Wonkam, A., Kengne, A.P., Ntsekhe, M. and Dandara, C., 2019. Warfarin dose and CYP2C gene cluster: An African Ancestral-specific variant is a strong predictor of dose in Black South African Patients. *Omics: a journal of integrative biology*, *23*(1), pp.36-44. **2.37**
55. Hamda C, Sangeda R, Mwita L, Meintjes A, Nkya S, Panji S, Mulder N, Guizani-Tabbane L, Benkahla A, Makani J, Ghedira K; **Chimusa, E** ,H3ABioNet Consortium. A common molecular signature of patients with sickle cell disease revealed by microarray meta-analysis and a genome-wide association study. PLoS One. 2018 Jul 6;13(7):e0199461. doi: 10.1371/journal.pone.0199461. PMID: 29979707; PMCID: PMC6034806. **3.24**

## 2017

56. Choudhury, A., Ramsay, M., Hazelhurst, S., Aron, S., Bardien, S., Botha, G., **Chimusa, E.R**., Christoffels, A., Gamieldien, J., Sefid-Dashti, M.J. and Joubert, F., 2017. Whole-genome sequencing for an enhanced understanding of genetic variation among South Africans. *Nature communications*, *8*(1), pp.1-12. **14.92**

57. Mugo, J.W., Geza, E., Defo, J., Elsheikh, S.S., Mazandu, G.K., Mulder, N.J. and **Chimusa, E.R**., 2017. A multi-scenario genome-wide medical population genetics simulation framework. *Bioinformatics*, *33*(19), pp.2995-3002. **6.937**

58. Mazandu, G.K., **Chimusa, E.R**., Rutherford, K., Zekeng, E.G., Gebremariam, Z.Z., Onifade, M.Y. and Mulder, N.J., 2018. Large-scale data-driven integrative framework for extracting essential targets and processes from disease-associated gene data sets. *Briefings in bioinformatics*, *19*(6), pp.1141-1152. **13.99**

59. Victor Jongeneel, Ph.D.; Ovokeraye Achinike-Oduaran; Ezekiel Adebiyi; Marion Adebiyi; Seun Adeyemi; Bola Akanle; Shaun Aron; Efejiro Ashano; Hocine Bendou; Gerrit Botha; **Emile R. Chimusa**; Ananyo Chowdhury; Ravikiran Donthu; Jenny Drnevich; Oluwadamila Falola; Christopher J Fields; Scott Hazelhurst; Liesl Mary Hendry; Itunuoluwa Isewon; Radhika S Khetani; Judit Kumuthini; Magambo Phillip Kimuda; Lerato Magosi; Liudmila Sergeevna Mainzer; Suresh Maslamoney; Mamana Mbiyavanga; Ayton Meintjes; Danny Mugutso; Phelelani Mpangase; Richard Munthali; Victoria Nembaware; Andrew Ndhlovu; Trust Odia; Adaobi Okafor; Olaleye Oladipo; Sumir Panji; Venesa Pillay; Gloria Rendon; Dhriti Sengupta; Nicola Mulder (2017) Assessing Computational Genomics Skills: Our Experience in the H3ABioNet African Bioinformatics Network Short title: H3ABioNet Node Assessment Exercise. PLoS computational biology, 13(6), p.e1005419. **4.428**

60. Geard, A., Pule, G.D., Chetcha Chemegni, B., Ngo Bitoungui, V.J., Kengne, A.P., **Chimusa, E.R**. and Wonkam, A., 2017. Clinical and genetic predictors of renal dysfunctions in sickle cell anaemia in Cameroon. *British journal of haematology*, *178*(4), pp.629-639. **5.67**

61. Nicola J. Mulder, Ezekiel Adebiyi, Marion Adebiyi, Seun Adeyemi, Azza Ahmed, Rehab Ahmed, Bola Akanle, Mohamed Alibi, Don L. Armstrong, Shaun Aron, Efejiro Ashano, Shakuntala Baichoo, Alia Benkahla, David K. Brown, **Emile R. Chimusa**, Faisal M. Fadlelmola, Dare Falola, Segun Fatumo, Kais Ghedira, Amel Ghouila, Scott Hazelhurst, Itunuoluwa Isewon, Segun Jung, Samar Kamal Kassim, Jonathan K. Kayondo, Mamana Mbiyavanga, Ayton Meintjes, Somia Mohammed, Abayomi Mosaku, Ahmed Moussa, Mustafa Muhammd, Zahra Mungloo-Dilmohamud, Oyekanmi Nashiru, Trust Odia, Adaobi Okafor, Olaleye Oladipo, Victor Osamor, Jellili Oyelade, Khalid Sadki, Samson Pandam Salifu, Jumoke Soyemi, Sumir Panji, Fouzia Radouani, Oussama Souiai, Özlem Tastan Bishop., 2017. Development of bioinformatics infrastructure for genomics research. *Global heart*, *12*(2), pp.91-98. **3.426**

62. Lebeko, K., Manyisa, N., **Chimusa, E.R**., Mulder, N., Dandara, C. and Wonkam, A., 2017. A genomic and protein–protein interaction analyses of nonsyndromic hearing impairment in Cameroon using targeted genomic enrichment and massively parallel sequencing. *Omics: a journal of integrative biology*, *21*(2), pp.90-99. **2.37**

63. Yocgo, R.E., Geza, E., **Chimusa, E.R**. and Mazandu, G.K., 2017. A post-gene silencing bioinformatics protocol for plant-defence gene validation and underlying process identification: case study of the Arabidopsis thaliana NPR1. *BMC plant biology*, *17*(1), pp.1-18. **4.215**

64. Pule, G.D., **Chimusa, E.R**., Mnika, K., Mhandire, K., Kampira, E., Dandara, C. and Wonkam, A., 2017. Beta-globin gene haplotypes and selected Malaria-associated variants among black Southern African populations. *Global health, epidemiology and genomics*, *2*. **2.407**

65. Dandara, C., **Chimusa, E.R**. and Wonkam, A., 2017. South African student protests, 2015-2016: The aftermath viewed through Medical Science Honours students at the University of Cape Town. *South African Medical Journal*, *107*(9), pp.723-723. **1.325**

66. Mazandu, G.K*., **Chimusa, E.R*.** and Mulder, N.J., 2017. Gene ontology semantic similarity tools: survey on features and challenges for biological knowledge discovery. *Briefings in bioinformatics*, *18*(5), pp.886-901. **13.99**

## 2016

67. Mazandu, G.K*., **Chimusa, E.R*.,** Mbiyavanga, M. and Mulder, N.J., 2016. A-DaGO-Fun: an adaptable Gene Ontology semantic similarity-based functional analysis tool. *Bioinformatics*, *32*(3), pp.477-479. **6.937**
68. Mulder, N.J., Adebiyi, E., Alami, R., Benkahla, A., Brandful, J., Doumbia, S., Everett, D., Fadlelmola, F.M., Gaboun, F., Gaseitsiwe, S. and Ghazal, H., **Emile R. Chimusa** (2016). H3ABioNet, a sustainable pan-African bioinformatics network for human heredity and health in Africa. Genome research, 26(2), pp.271-277. **9.043**
69. Mazandu, G.K†*., **Chimusa, E.R†*.,** Mbiyavanga, M., Mulder, N.J. and Mazandu, M.G.K., (2016). The A-DaGO-Fun Package. Bioinformatics, 32(3b), pp.474-479. **6.937**
70. Mazandu, G.K*., **Chimusa, E.R*.** and Mulder, N.J., 2016. Expressions of different semantic similarity measures in the context of biomedical sciences and WordNet. *Brief Bioinform*, *18*(5), pp.886-901. **13.99**

## 2015

71. **Chimusa, E.R*.,** Mbiyavanga, M., Mazandu, G.K. and Mulder, N.J., 2016. ancGWAS: a post genome-wide association study method for interaction, pathway and ancestry analysis in homogeneous and admixed populations. *Bioinformatics*, *32*(4), pp.549-556. **6.937**
72. **Chimusa, E.R***., Mbiyavanga, M., Masilela, V. and Kumuthini, J., 2015. "Broadband" Bioinformatics skills transfer with the Knowledge Transfer Programme (KTP): educational model for upliftment and sustainable development. *PLoS computational biology*, *11*(11), p.e1004512. **4.428**
73. **Chimusa, E.R***., Meintjies, A., Tchanga, M., Mulder, N., Seoighe, C., Soodyall, H. and Ramesar, R., 2015. A genomic portrait of haplotype diversity and signatures of selection in indigenous southern African populations. *PLoS genetics*, *11*(3), p.e1005052. **5.917**
74. Kumuthini, J., Mbiyavanga, M., **Chimusa, E.R**., Pathak, J., Somervuo, P., Van Schaik, R.H., Dolzan, V., Mizzi, C., Kalideen, K., Ramesar, R.S. and Macek, M., 2016. Minimum information required for a DMET experiment reporting. *Pharmacogenomics*, *17*(14), pp.1533-1545. **3.55**

## 2014

75. **Chimusa, E.R*.,** Zaitlen, N., Daya, M., Möller, M., van Helden, P.D., Mulder, N.J., Price, A.L. and Hoal, E.G., 2014. Genome-wide association study of ancestry-specific TB risk in the South African Coloured population. *Human molecular genetics*, *23*(3), pp.796-809. **6.15**
76. Kumuthini, J., **Chimusa, E**. and Masilela, V., 2014. Bioinformatics Knowledge Transfer Programme (KTP) for Human and Capital Development in South Africa. *EMBnet. journal*, *20*(A), p.791. **1.71**

## 2013

77. Daya, M., Van Der Merwe, L., Galal, U., Möller, M., Salie, M., **Chimusa, E.R**., Galanter, J.M., Van Helden, P.D., Henn, B.M., Gignoux, C.R. and Hoal, E., 2013. A panel of ancestry informative markers for the complex five-way admixed South African coloured population. *PloS one*, *8*(12), p.e82224. **3.24**
78. **Chimusa, E.R*.** 2013. Mapping genes underlying ethnic differences in tuberculosis risk by linkage disequilibrium in the South African coloured population of the Western Cape.

79. **Chimusa, E.R*.,** Daya, M., Möller, M., Ramesar, R., Henn, B.M., Van Helden, P.D., Mulder, N.J. and Hoal, E.G., 2013. Determining ancestry proportions in complex admixture scenarios in South Africa using a novel proxy ancestry selection method. *PloS one*, *8*(9), p.e73971. **3.24**
80. De Wit, E., Delport, W., **Chimusa, E.R.,** Meintjes, A., Möller, M., Van Helden, P.D., Seoighe, C. and Hoal, E.G., 2010. Genome-wide analysis of the structure of the South African Coloured Population in the Western Cape. *Human genetics*, *128*(2), pp.145-153. **5.2**

***CHAPTERS IN BOOKS***

1. Mazandu, G.K., Geza, E., Seuneu, M. and **Chimusa, E.R*., 2019**. Orienting Future Trends in Local Ancestry Deconvolution Models to Optimally Decipher Admixed Individual Genome Variations. In Bioinformatics Tools for Detection and Clinical Interpretation of Genomic Variations. IntechOpen. DOI: 10.5772/intechopen.8276.
2. Mazandu, G.K., Kyomugisha, I., Geza, E., Seuneu, M., Bah, B. and **Chimusa, E.R*., 2019**. Designing Data-Driven Learning Algorithms: A Necessity to Ensure Effective Post-Genomic Medicine and Biomedical Research. In Machine Learning in Medicine and Biology. IntechOpen. DOI: 10.5772/intechopen.84148.
3. Elsheikh, S.S., Bakas, S., Mulder, N.J., **Chimusa, E.R**., Davatzikos, C. and Crimi, A., 2018, September. Multi-stage association analysis of glioblastoma gene expressions with texture and spatial patterns. In *International MICCAI Brainlesion Workshop* (pp. 239-250). Springer, Cham.
4. Kumuthini, J., Zass, L., **Chimusa, ER.**, Chaouch, M. and Masimiremwa, C., **2018**. Minimum Information Required for Pharmacogenomics Experiments. In Human Genome Informatics (pp. 179-193). Academic Press, doi.org/10.1016/B978-0-12-809414-3.00009-7.

***BOOKS***

5. Emile R. Chimusa* (2016) Mapping Genes Underlying Ethnic Differences in Admixed Populations. LAP LAM. Aca. Nov. 30, 2016. ISBN-13: 978-3330010734, 236 pages.

***2023 UNDER REVIEW MANUSCRIPT***

1. Yosr Haffani and **Chimusa ER***, 2023 Investigation of the association between genetic variations in key genes of estrogen signaling; metabolism and molecular immune checkpoints and breast cancer aggressiveness and response to different therapeutic modalities. Human molecular genetics.
2. Denis Awany, and **Chimusa ER***, 2023. Associating Microbiome with Host Phenotype using Dirichlet Process Random Effects Model. Molecular Biology and Evolution.
3. Mugo J and **Chimusa ER*,** 2021. JasMAP: A Joint Ancestry and SNP Association Method for Multi-way Admixed Population. Bioinformatics.
4. Iliana Boura, Gloudi Agenbag, **Emile R. Chimusa*,** 2023. In Silico Network Analysis Revealed Genetic Modifiers of Spondyloepimetaphyseal Dysplasia with Joint Laxity in South African Afrikaner Families. Frontier in Genetics.
5. Mugo J., Christian DB, Mulder MJ, and **Chimusa ER*,** 2023, Data Simulation to Optimize a Framework for Genome-Wide Association Studies in Diverse Populations. Frontiers in Genetics.
6. Wonderful T. Choga, Prisca K. Thami, Kabongo E. Ntumba, Simani Gaseitsiwe, **Chimusa ER***, 2023, Genome-wide selection-scans reveal signatures of polygenic adaptation in Batswana withIV-1 infection.

7. Wonderful T. Choga, Prisca K. Thami, Motswedi Anderson, Sikhulile Moyo, **Chimusa ER*,** Simani Gaseitsiwe*, 2023. Botswana and HIV-1 infection (1985–2020): frontiers, challenges and recommendations.
8. Wonderful T. Choga, Prisca K. Thami, Kabongo E. Ntumba, Simani Gaseitsiwe, **Chimusa ER*,** 2023. Current opinion on function Genome-wide Association studies (fGWAS) on HIV-1 clinical outcomes among Africa populations. Briefings in Bioinformatics.
9. Prisca K. Thami, Wonderful T. Choga, Delesa D. Mulisa, Collet Dandara, Andrey K. Shevchenko, Melvin M. Leteane, Vlad Novitsky, Stephen J. O'Brien, Myron Essex, Simani Gaseitsiwe, **Chimusa ER*,** 2023. Whole genome rare-variant association study of HIV-1 progression in a Southern African population.. https://doi.org/10.1101/2020.12.16.20248307. (Frontiers in Genetics/).
10. Prisca K. Thami, Wonderful T. Choga, Delesa D. Mulisa, Collet Dandara, Andrey K. Shevchenko, Melvin M. Leteane, Vlad Novitsky, Stephen J. O'Brien, Myron Essex, Simani Gaseitsiwe, Chimusa ER*. 2023. Whole genome sequencing-based characterization of human genome variation and mutation burden in Botswana.. http://dx.doi.org/10.1101/2020.12.15.422821.h (Human molecular genetics).

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## Teaching Experience, Vision and Philosophy

**Overview**

For the last 8 years, I have been program convener and responsible for bioinformatics training in Medical Sciences BSc Honours program, specializing in Human genetics or Forensics Honours at UCT Faculty of Health Sciences, consisting of several students across Africa and international. I am program director of an interdisciplinary honours, masters, and PhD curricula in Computational Health Informatics Program (CHIP) that I developed to introduce rigorous in data science, computational multi-omics, clinical informatics, and imaging data science for Biomedical graduate students. I secured for this program a research training grant "Computational Omics & Biomedical Health Informatics" (grant 1U2RTW012131-01) as PI from National Institution for Health (NIH, US) - Harnessing Data Science for Health Discovery and Innovation in Africa (DS-I Africa).

I have strong experiences in teaching graduate and postgraduate students in interdisciplinary programs and students from diverse genetics, culture, and religious backgrounds.

## Selected recent course modules

| *Course name* | *Brief Content* | *Class size* |
|---|---|---|
| Bioinformatics programming with python | ✓ *Covers advance programming with python to apply data science and analysis solutions to biological and medical situations and analyse raw data and present it effectively in a visualization.*<br>✓ *covers tools development to build and modify robust models, using techniques such as regularization, to help trainees solve* | *20-60* |

| | | |
|---|---|---|
| | *health and biological problems.*<br>✓ *Covers practical grounding in Jupyter Notebook, one of the most widely used notebooks in data science.* | |
| Machine learning and biomedical data science | ✓ *Covers computational statistical genomics, machine learning and deep-learning approaches to solve problems of various omics and related downstream omics data analysis, and how to turn biomedical data into information relevant to clinical application and public health research.* | *20-60* |
| Bioinformatics for Next Generation Sequencing | ✓ *Covers state-of-the art bioinformatics techniques in next generation sequence, gene expression data analysis and their underlying enrichment analysis.* | *40-60* |
| Medical populations genetics | ✓ *Covers deep principles of medical population genetics and evolutionary forces.*<br>✓ *Covers various approaches used in clustering-based population genomics.*<br>✓ *Covers statistical Omics to analysis evolutionary events and mechanism of changes of biological variants relevant to health.* | *35-70* |
| Genome-Wide Association Studies | ✓ *Covers association-based machine learning, polygenic risk scores, cross-metanalysis and principles of various Omics-phenotype association studies, including discussion of their statistical approaches and study cases, study design, power calculation and clinical implication.* | *35 - 70* |

My teachings align with several aspects of ***Innovation in teaching and learning.*** All my teachings include the following:

- Limiting didactic lectures to fundamental concepts and principles.
- Considering case/problem-based learning approach – use of current/recent issues and real-world problems for teaching.
- Usage of technology to enhance learning in the classroom (e.g., eLearning Portal system and iPhone, Panopto, PowerPoint to record, live-stream etc).
- Flipping classroom (students prepare material before class, to enable deeper discussion on the topics during class time).
- Supplementing teaching time with online resources (videos, articles).
- Introducing debates on controversial and current issues (e.g., missing heritability in a model of risk prediction, relevancy of integrative multi-omics for a great understanding of system biology etc.).
- Consideration of showcasing a range of science-related careers (including non-academic careers).
- Allowing students group work and presentations, engagement with students, and use of feedback on draft assignments to foster student reflection to possibly address cultural diversity.

**Demonstrating Standards in Teaching**

- In my teaching, I promote self-learning and help students develop critical thinking skills and retain knowledge that leads to self-actualization. I frequently incorporate feedback from students to improve my courses and other courses under my program convenorship.
- I have incorporated into my teaching the usage of multimedia in advanced MS PowerPoint, MS Team, and talks on new technology to allow diversity in learning.

**Addressing cultural diversity in teaching and academic progress of students**

- I have initiated regular interaction/engagement to build relationships with students, identify their task orientation, capture their academic progress, and receive frequent feedback.

**Student mentor with demonstrated ability to adapt approaches and assist students with a wide variety of non-academic needs**

**Applying criteria for assessment practice and educational literature**

- In the past 10 years, I have regularly been invited to act as an external examiner at national and international universities to examine PhD theses and master's dissertations.
- I have been on the exam committee and Examinations/Recruitment Committee from my previous faculties.
- I have incorporated alignment, consistency, security, transparency, and fairness approaches in the assessment of modules/courses taught as part of my role at my current and past universities.

**Scholarship**

- I have experience in developing new Bioinformatics and omics data science curricula (Chimusa et al. 2015 PLoS Comput Biol. PMID: 26583922; Mulder et al. 2017 Global Heart PMID: 28302555; Jongeneel et al. 2017 PLoS Comput Biol. PMID: 28570565), course materials and training in Bioinformatics, Statistical Genomics, advanced in Bio-computing with R/Python/shell computer programs, medical population genetics, Genome-wide Associations Studies and Next Generation Sequencing.
- I published peer-reviewed education articles (Chimusa et al. 2015; Dandara et al. 2017; Jongeneel et al. 2017; Kumuthini et al. 2014) to support efficient best practices and usage of technology strategies in teaching and research training in high education.

**Initiating and introducing new courses and/or Curriculum**

Leveraging my experiences in teaching and developing new curricula, I will collaborate with local researchers at Leeds Beckett University to develop/adapt, evaluate, and sustain new or existing interdisciplinary graduate training programme in omics data science to cultivate researchers and leaders in biomedical data science focused on the needs and global and local priorities through machine learning and computational paradigms for genomics and imaging data and their application for the development of innovative solutions.

My vision of that new or adapted training program is to harness the value of genomic data science and biomedical informatics to improve clinical and translational research and bridge the gap between academia and industry.

## Research Statement

In the past 10 years, I tackled relevant health challenges by developing Bioinformatics tools (https://github.com/echimusa) for functional genomics, post-genome-wide Association Studies (GWAS), disease risk prediction, Gene Ontology semantic similarity that are widely in the Genomics and Bioinformatic research communities. These tools provide insights into deconvoluting the pathogenesis of complex diseases.

I hereby cite only four of them

(1) Detecting and characterizing population structure is important for a wide range of applications, including correcting confounding due to population stratification and identifying signals of natural selection. Inaccurate parental populations for admixed populations can result in erroneously inferred ancestry and affect the detection of disease-gene with substantial variation in susceptibility. I have addressed these challenges by developing "PROXYANC" tool to account for inaccurate parental populations for admixed populations (Chimusa et al., 2013, PMID: 24066090).

(2) Genome-wide Association studies may miss disease genes with weak genetic effects or strong epistatic effects due to the single-marker testing approach commonly used. I addressed this challenge, by introducing a post-GWAS tool "ancGWAS", an algebraic graph-based centrality measure that leverages the human protein-protein interaction information, local ancestry (in the case of admixed populations), and linkage disequilibrium data to mine GWAS summary statistics in deconvoluting the interactions between genes and identifying significant disease sub-networks/pathways underlying the pathogenesis of complex diseases (Chimusa et al, 2016, PMID: 26508762).

(3) "FractalSIM", designed under the whole genome resampling simulation approaches, to mimic various scenarios of population genetic diversity, genomic variation affecting diseases and DNA sequence patterns of admixed and/or homogeneous populations, particularly those under natural selection pre- and post-admixture process models. It offers broad users the option to simulate various scenarios of population bottlenecks, complex admixture, selection, and disease models. It has advantages over existing tools: (i) it integrates several genetics simulation models into one framework and (ii) it can simulate genome-wide data (Mugo et al. 2017, PMID: 28957497).

(4) A-DaGO-Fun (ADaptable Gene Ontology semantic similarity based Functional analysis), a web application software. It is a portable software package integrating all known GO information content-based semantic similarity measures and relevant biological applications associated with these measures. A-DaGO-Fun has the advantage not only of handling datasets from the current high-throughput genome-wide applications but also allowing users to choose the most relevant semantic similarity approach for their biological applications and to adapt a given module to their needs (Mazandu et al., 2016, PMID: 26476781).

Large genome-wide association studies (GWAS) have been crucial in detecting disease-associated variants in recent years. I have contributed to the field by both developing novel methods as well as performing GWAS in African structured and admixed populations by presenting a broad survey of polymorphisms in a novel array genotyping, sequencing, and copy number of polymorphism data sets of self-identified ethnic/linguistic populations of Africa (Esoh et al. 2021 Sci Rep PMID: 33441574; Choudhury et al., 2020 Nature. PMID: 33116287; Choudhury et al., 2017 Nat Commun. PMID: 29233967; Chimusa et al., 2015 PLoS Genet. PMID: 25811879; Chimusa et al. 2013 PLoS One. PMID: 24066090; de Wit et. al. 2010 Hum Genet. 2010 PMID: 20490549).

I have made significant research contributions to dissecting genetics risk scores and highlight their research and clinical potential (Damena et al. 2021 Front Genet. PMID: 34868193; Damena et al. 2020 Hum Mol Genet. PMID: 31691794; Wonkam et al. 2020 Clin Transl Med PMID: 32898326; Agamah et al 2019 Brief Bioinform, PMID: 31711157;Awany et al. 2018, Brief Funct Genomics, PMID: 30605512; Chimusa et al. 2014; Hum Mol Genet. PMID: 2405767). Our development of data standards, gene ontology and Sickle Cell Diseases Ontology are significant contributions to the field and are being used for harmonizing data and clinical decision-making (Mazandu et al. 2018 Brief Bioinform PMID: 28520909; Sickle Cell Disease Ontology Working Group et al. 2019 Database (Oxford) PMID: 31769834; Mazandu et al, 2017 Brief Bioinform PMID: 27473066; Mazandu et al. PMID: 26476781).

I am the principal/co-investigator of several collaborative NIH-funded projects and consortia, particularly the funded H3ABioNet (U24HG006941) a pan-African bioinformatics network for H3Africa consisting of 28 institutions in 16 African countries. Through these networks, I contributed to the development of standards for phenotype data for cohort studies, with associated data dictionaries, African genome variation database as well as data harmonization across multiple projects, and the design of the first Africa-specific genotyping chip/array (H3A chip). I have implemented workflows for genomic data analysis and implemented FAIR principles to facilitate reproducible science.

I have experience in developing new Bioinformatics and genomics data science curricula (Chimusa et al. 2015 PLoS Comput Biol. PMID: 26583922; Mulder et al. 2017 Global Heart PMID: 28302555;

Jongeneel et al. 2017 PLoS Comput Biol. PMID: 28570565), course materials and training in Bioinformatics, Statistical Genetics, advanced in Bio-computing with R/Python/shell computer programs, medical population genetics, Genome-wide Associations Studies and Next Generation Sequencing. I have been a program convener and responsible for Bioinformatics training in Medical Sciences BSc honors, Masters, and PhD programs at UCT Faculty of Health Sciences, consisting of several students across Africa and internationally. Under my supervision, I successfully graduated with several PhDs and master's students and a postdoctoral researcher fellow who became an associate professor.

**Short and longer-term vision of research and strategic goals**

The delineation of health and disease from association mapping holds promise to bridge the gap between statistical genomics and clinical translation, thereby improving diagnostics, screening, genetic testing, and counseling in global clinical populations. It is widely recognized that health systems must put more emphasis on prevention and adopt a person-centered approach. Artificial Intelligence (AI) and Machine Learning (ML) approaches along with the increased availability of genomics and health data hold great potential to pave the way for personalised prevention and enable progress toward disease risk prediction and early detection of complex diseases. In addition, these approaches provide unprecedented opportunities to catalogue the spectrum of individual variation in traits, treatments, and responses to drugs, to isolate disease-causing variants and causative mutations relevant for diagnosis and therapy. However, the exceptional polygenicity of human traits makes unraveling mechanisms from current disease-gene association mapping and risk prediction daunting.

Over the past 10 years, my research focused on the development of computational (AI/ML) method tools pertinent to genomics and environmental determinants of susceptibility to and risk of complex diseases, genome variation, microbial genomics, and infectious diseases from both the host and pathogen perspectives to facilitate the transformation of genomics-driven clinical practice.

I am placed to make contributions to ongoing research at Leeds Beckett University. My short research plan and goals, will be to extend and elaborate on my current research from integrative genomics and/or data analytics-driven approaches for complex disease risk prediction to:

1. Foster multidisciplinary collaborative research and communication links between existing research groups within Leeds Beckett University.
2. Leverage existing high-quality health-relevant data for the rigorous development of association analyses and the AI disease-risk prediction tools in the spectrum of non-communicable and communicable traits.
3. Contribute to the development of new and improved non-communicable and communicable diseases digital biobank, deep-learning driven natural language processing approaches to advance genomics-driven healthcare.
4. Transferring of data science tools and computational methods that will facilitate the transformation of host-pathogen genomics-driven clinical practice.
5. Design statistical models for leveraging (a) cross-population gene/sub-networks/pathways meta-analysis to recover disease association signal risk that could be missed from single-marker association tests such as from Genome-wide Association studies, (b) cross-population gene/sub-networks/pathways or pathways drug responses and regulations in homogeneous or recently admixed populations.
6. Establish genomics-driven host-pathogen interactome by developing a systems-level based integrative multi-omics methodology to investigate functional components and the underlying mechanisms associated with complex communicable/non-communicable disease states, examine risk factors, and predict druggable proteins, and existing drugs that target host factors.
7. Attract national and international research funding and contribute to greater investment in genomics and predictive medicine.

8. Develop, support, and contribute to capacity development will be achieved through training and promoting interactivity between researchers across disciplines within Leeds Beckett University and;
9. Build on broad stakeholder engagement, and support proposals developing novel robust, and trustworthy AI tools to enable timely personalised prevention approaches for complex diseases.

**My main vision is the development of AI/ML methodologies and approaches to improving genomics risk assessment, risk prediction of complex diseases, and advanced genomic medicine**. From the long-term plan, I will expand on the above short-term research plan to:

1. Generate new high-quality health data (i.e. cohorts, electronic health records, and registries, considering the individual's genotypic/phenotypic, medical, lifestyle, socioeconomic, behavioural data etc.);
2. Continue contributing to the development of new and improved existing digital biobanks within the Leeds Beckett University, and deep-learning-driven natural language data processing approaches to advance genomics-driven health care.
3. Develop AI tools for assessing and predicting individuals' risk of developing a disease and/or the risk of disease progression once it is diagnosed, considering the individuals' genotypes, phenotypes, family history of the disease, lifestyle, occupational/environmental stressors and/or socio-economic, ancestry effects and behavioural characteristics.
4. Develop integrative multi-omics approaches to allow benchmarking disease risk and guiding the development of host-directed therapies, novel surveillance, diagnostics, treatment, and prevention tools.
5. Leverage my experience in Small and medium-sized enterprises (SME) to strengthen the scientific and technological basis of SME(s) and valorise their innovations for Leeds Beckett University's benefit.
6. Engage in partnerships with various local and international bioscience/biotechnology industries to (a) stimulate inter-disciplinary thinking and career ambitions, (b) initiate professional networks to inform and facilitate career choices for our trainees and mainly obtain support and secure internships for trainees and (c) encourage and participate to greater investment in genomics and predictive medicine.

Globally, there is an increasing need for (a) integrative, innovative, and cost-effective genomics data science infrastructure for data management and analyses which enables and promotes cutting-edge translational research; (b) building predictive disease models to identify novel mechanisms for disease prevention, control, diagnosis, and treatment outcome prediction. This research is extremely relevant to the local setting and addresses very important global health issues, particularly it covers important and current key strategic focus areas of The National Institute for Health Research (NIHR), UK and Horizon Europe 2021 - 2030 - Horizon Europe program, namely AI/ML and Health.

I believe, the proposed plan will contribute to the generalizability and transferability of AI disease risk predictive tools across diverse populations and decrease the current disparity in predictive and genomics medicine.
The proposed research plan will benefit from (a) the established collaboration with other existing research groups within Leeds Beckett University, and (b) the genomics and computational resources from Leeds Beckett University. I anticipate that my research and teaching will have decisive impacts, advance, and build capacity for genomics data science through the development and transfer of developed tools and computational methods that will facilitate the transformation of genomics-driven clinical practice.

## References


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